

Use Cases of Image Point Processing Operations

Practical 1: Image Negation

Use Case 1: Medical Imaging (X-ray Analysis)

Problem Statement:

Radiologists often examine the **negative of an X-ray image** because bones, tissues, and abnormalities may become more distinguishable in the inverted image. Write a program to read a grayscale X-ray image, generate its negative, and compare the original and negative images.

Use Case 2: Satellite Image Analysis

Problem Statement:

A satellite image contains dark roads on a bright background. To improve visual interpretation, generate the negative of the image so that roads appear brighter than the surrounding area.

Use Case 3: Historical Document Enhancement

Problem Statement:

Old handwritten manuscripts often contain faded dark text on light paper. Generate the negative image to improve the visibility of the handwritten content.

Practical 2: Log Transformation

Use Case 1: Dark Image Enhancement

Problem Statement:

A surveillance camera captures images under poor lighting conditions, making dark regions difficult to observe. Apply a logarithmic transformation to enhance the darker portions of the image while compressing brighter regions.

Use Case 2: Astronomical Image Processing

Problem Statement:

An astronomical image contains faint stars that are barely visible due to low intensity values. Apply logarithmic transformation to reveal the dim stars without excessively brightening already bright objects.

Use Case 3: Medical Imaging

Problem Statement:

A CT scan contains low-intensity anatomical structures that are difficult to identify. Enhance these structures using logarithmic transformation and compare the result with the original image.

Practical 3: Gamma (Power-Law) Transformation

Use Case 1: Correcting Underexposed Images

Problem Statement:

A photograph captured in a poorly lit environment appears too dark. Apply Gamma Correction ($\gamma < 1$) to improve its brightness while preserving image details.

Use Case 2: Correcting Overexposed Images

Problem Statement:

An outdoor photograph captured under strong sunlight appears excessively bright. Apply Gamma Correction ($\gamma > 1$) to reduce brightness and improve visibility.

Use Case 3: Display Calibration

Problem Statement:

Different display devices (monitors, TVs, projectors) reproduce brightness differently. Apply gamma correction with different gamma values (0.5, 1.0, 1.5, and 2.2) to observe their effect on image appearance.

Practical 4: Comparison of Intensity Transformations

Use Case: Image Enhancement for Different Applications

Problem Statement:

An image enhancement software offers multiple intensity transformation techniques. Develop a program that applies **Image Negation**, **Log Transformation**, and **Gamma Transformation** to the same image. Compare the visual results and identify the most suitable transformation for:

- Medical X-ray images
- Low-light photographs
- Overexposed images